

UndoProof

Prove the way back

PostgreSQL rollback rehearsal by Shivam Gupta

The rollback succeeds. The new order disappears.

A release can pass forward tests while its recovery path breaks the previous application contract or erases data written after deployment. UndoProof makes that failure reproducible before a migration ships.

HOW IT WORKS

A release contract supplies synthetic seed data, migration and rollback SQL, old and new query contracts, representative writes, and a canonical data invariant. UndoProof executes isolated PostgreSQL rehearsals through PGLite. It shows the actual SQL, errors, and before-and-after rows, then exports evidence and a focused IBM Bob repair prompt.

THE IMPORTANT CHECK

Write a new order *after* the migration. Roll back. Compare the customer-visible data. A safe recovery can retain a compatible expanded schema when the application rolls back, instead of restoring an obsolete snapshot.

FOR DEVELOPERS

Review the failed release state, inspect a repair, and rerun the same contract. The browser and command-line workflow share an execution engine. Rehearsals need no production credentials or paid model inference.

COMMERCIAL HYPOTHESIS

Free MIT local runner. A proposed \$99/month team workspace would add shared policies and retained evidence. Initial audience: platform leads at SaaS teams. Pricing and demand require customer validation.

SCOPE

Results cover the supplied SQL and fixtures in embedded PostgreSQL. They do not establish full application behavior, concurrent transaction behavior, production lock safety, or production parity.

shi1720.github.io/ibm-bob

Source: github.com/shi1720/ibm-bob

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